LESSON 6.4

# Write and Evaluate Numerical Expressions with Exponents

**Start here** · Words, worked example, and sentence starters

**Name** _____ **Date** _____ **Period** _____

**LEARNING TARGET** I can write numerical expressions from a real situation and evaluate them using the order of operations, including powers.

**WHY THIS MATTERS** The studio's booking sheet lists a price for each kind of seat. Writing one numerical expression for the whole order — and evaluating it in the right order — is faster and easier to check than adding costs one at a time. Some counts are written as powers, so the exponent has to be handled before any multiplying. The booking sheet lists a $150 base fee plus $12 for each speaker rented. Write one expression for the cost of renting 5 speakers.

START HERE

## Words for this lesson

Read each word before you begin. Say it out loud.

| WORD | WHAT IT MEANS | EXAMPLE |
|---|---|---|
| **Numerical expression** *Spanish: Expresión numérica* | A math phrase built from numbers and operations, with no equal sign and no variable. | 4 × 9 + 3 × 6 |
| **Evaluate** *Spanish: Evaluar* | To find the single value an expression is worth by carrying out its operations in order. | 3 × 6 = 18 |
| **Order of operations** *Spanish: Orden de las operaciones* | The agreed order for evaluating: grouping symbols, then exponents, then multiply and divide left to right, then add and subtract left to right. | 3 + 4 × $2^2$ → 3 + 4 × 4 → 3 + 16 = 19 |
| **Power** *Spanish: Potencia* | A short way to write repeated multiplication of the same factor, such as $2^3$ for 2 × 2 × 2. | 8 × 8 = 8^2 |
| **Base** *Spanish: Base* | In a power, the number being multiplied by itself. In $2^3$ the base is 2. | In 8^2, the base is 8. |
| **Exponent** *Spanish: Exponente* | In a power, the small raised number that says how many times the base is used as a factor. | In 8^2, the exponent is 2. |

HOW IT WORKS

## Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

**Ms. Ortiz buys 4 pizzas at $9 each and one $6 salad. Write an expression for the total. Evaluate it.**

1. Pizzas are equal groups: 4 × 9.
2. Add the salad: 4 × 9 + 6.
3. Evaluate: 36 + 6 = 42.

**Answer:** 4 × 9 + 6 = $42

NOW YOU TRY

## Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

**Suppose the student tickets cost $2^3$ dollars each instead of $2. The expression becomes $2 \times 15 + 4 \times 20 + 8 \times 2^3$.**

1. _____
2. _____
3. _____

**Answer:** _____

SAY IT AND WRITE IT

## Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- The word _____ tells me to multiply _____ by _____ .
- The expression for this situation is _____ .
- I used a power because the same factor repeats _____ times.
- An expression has no equal sign, so I only _____ it.

**Word bank** numerical expression | evaluate | power | base | exponent | product | sum | difference | each | in all | operation | value

WATCH OUT!

## A common mistake

A common mistake with powers is multiplying the base by the exponent: reading $3^2$ as $3 \times 2 = 6$ instead of $3 \times 3 = 9$. A second is evaluating strictly left to right, so $3 + 4 \times 2$ becomes 14 instead of 11. The order of operations exists so one expression has exactly one value: grouping symbols, then powers, then × and ÷ left to right, then + and − left to right.

LESSON 6.4

# Write and Evaluate Numerical Expressions with Exponents

**Version A** · Supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

**Name** _____ **Date** _____ **Period** _____

**LEARNING TARGET** I can write numerical expressions from a real situation and evaluate them using the order of operations, including powers.

**WHY THIS MATTERS** The studio's booking sheet lists a price for each kind of seat. Writing one numerical expression for the whole order — and evaluating it in the right order — is faster and easier to check than adding costs one at a time. Some counts are written as powers, so the exponent has to be handled before any multiplying. The booking sheet lists a $150 base fee plus $12 for each speaker rented. Write one expression for the cost of renting 5 speakers.

## PART 1 Understand the idea

models and meaning

**1** **Circle the letter of the best answer. Show how you know.**

What does $2^3$ mean?

(A) $2 \times 2 \times 2$
(B) $2 \times 3$
(C) $2 + 2 + 2$
(D) $3 \times 3$

**HINT** Which number is the base, and which is the exponent?

SHOW YOUR WORK

**2** **Circle the letter of the best answer. Show how you know.**

Sofia buys 4 tickets at $20 each and 2 tickets at $15 each. Which expression shows the total?

(A) $4 \times 20 + 2 \times 15$
(B) $4 + 20 + 2 + 15$
(C) $(4 + 2) \times (20 + 15)$
(D) $4 \times 2 + 20 \times 15$

**HINT** What does "each" tell you to do?

SHOW YOUR WORK

**3** **Complete the table. Show how you found each value.**

Evaluate each expression, following the order of operations.

| Expression | Value |
|---|---|
| $2 + 3 \times 4$ | 14 |
| $10 - 2^2$ | |
| $(2 + 3) \times 4$ | |

**HINT** For $10 - 2^2$: evaluate $2^2 = 4$ first.

**SHOW YOUR WORK**

## PART 2 Practice the skill

show every step

**4** **Circle the letter of the best answer. Show how you know.**

Evaluate $3 + 4 \times 2$.

(A) 9

(B) 11

(C) 14

(D) 24

**HINT** Which operation comes first in the order of operations?

**SHOW EACH PARTIAL PRODUCT, THEN ADD THEM**

1. **Split** *Break one factor apart.*
2. **Multiply** *One part at a time.*
3. **Add** *Add the parts.*

## PART 3 Apply it

real-world problems

**5** **Circle the letter of the best answer. Show how you know.**

Evaluate $5^2 + 3$.

(A) 13

(B) 16

(C) 28

(D) 64

**HINT** What must you evaluate first?

| Rewrite the whole expression each line | What I simplified |
|---|---|
| | |
| | |
| | |
| | |
| | |

1. **Grouping** *Inside ( ) first.*
2. **Exponents** *Then powers.*
3. **× ÷** *Left to right.*
4. **+ −** *Left to right.*

## PART 4 Explain and correct

reasoning

**6** **Find the mistake. Explain it, then show the correct work.**

**Find the error**

1. **Expression** Evaluate $3 + 4 \times 2^2$
2. **Step 1** $3 + 4 = 7$
3. **Step 2** $7 \times 2^2 = 7 \times 4 = 28$

**Which step has the mistake? Explain what went wrong, then show the correct work.**

**HINT** Which stage comes first: adding or the power?

CORRECT WORK

**Correct answer:** _____

EXPLAIN YOUR THINKING

**Two students evaluate 3 + 4 × 2 and get 14 and 11. Which one followed the order of operations, and why does the order have to be agreed on at all?**

SENTENCE STARTER The correct value is ___ because ___ comes before ___. Without an agreed order, the same expression would have ___ values.

**3-Point Exit Checklist:** ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

**How did it go?** ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

LESSON 6.4

# Write and Evaluate Numerical Expressions with Exponents

**Version B** · Core practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

**Name** _____ **Date** _____ **Period** _____

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**WHY THIS MATTERS** The studio's booking sheet lists a price for each kind of seat. Writing one numerical expression for the whole order — and evaluating it in the right order — is faster and easier to check than adding costs one at a time. Some counts are written as powers, so the exponent has to be handled before any multiplying. The booking sheet lists a $150 base fee plus $12 for each speaker rented. Write one expression for the cost of renting 5 speakers.

## PART 1 Understand the idea

models and meaning

**1** **Complete the table. Show how you found each value.**

Sofia buys 2 member tickets ($15), 4 non-member tickets ($20), and 8 student tickets ($2^3$ dollars). Build the total.

| Ticket type | Expression | Cost |
|---|---|---|
| Member | 2 × 15 | 30 |
| Non-member | 4 × 20 | |
| Student | 8 × $2^3$ | |

SHOW YOUR WORK

**2** **Write the name of the correct group on each line.**

Sort each item into the stage of the order of operations where it is handled.

**Groups:** Grouping symbols | Exponents | Multiply / divide | Add / subtract

_____ ( )

_____ $2^3$

_____ × and ÷ left to right

_____ + and − left to right

_____ [ ]

_____ $5^2$

## PART 2 Apply it

real-world problems

**3** **Circle the letter of the best answer. Show how you know.**

What is the total of $2 \times 15 + 4 \times 20 + 8 \times 2^3$?

(A) 126

(B) 174

(C) 158

(D) 1,392

SHOW EACH PARTIAL PRODUCT, THEN ADD THEM

**4** **Circle the letter of the best answer. Show how you know.**

Miguel evaluates $12 - 2 \times 3$ and gets 30. Margaret gets 6. Who is right?

(A) Miguel, because he worked left to right

(B) Both are right

(C) Neither is right

(D) Margaret, because multiplication comes before subtraction

SHOW YOUR WORK

## PART 3 Explain and correct

reasoning

**5** **Find the mistake. Explain it, then show the correct work.**

**Find the error**

1. **Expression** Evaluate $20 - 3^2$
2. **Step 1** $3^2 = 6$
3. **Step 2** $20 - 6 = 14$

**Which step has the mistake? Explain what went wrong, then show the correct work.**

CORRECT WORK

**Correct answer:** _____

EXPLAIN YOUR THINKING

**Two students evaluate 3 + 4 × 2 and get 14 and 11. Which one followed the order of operations, and why does the order have to be agreed on at all?**

**3-Point Exit Checklist:** ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

**How did it go?** ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

LESSON 6.4

# Write and Evaluate Numerical Expressions with Exponents

**Challenge** · Extension

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

**Name** _____ **Date** _____ **Period** _____

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## PART 1 Understand the idea

models and meaning

**1** **Complete the table. Show how you found each value.**

Evaluate each expression. Watch the stage each one tests.

| Expression | Value |
|---|---|
| $(4 + 6) \div 2$ | 5 |
| $4 + 6 \div 2$ | |
| $2 \times 3^2$ | |
| $(2 \times 3)^2$ | |

SHOW YOUR WORK

## PART 2 Explain and correct

reasoning

**2** Find the mistake. Explain it, then show the correct work.

### Find the error

1. **Expression** Evaluate $4 + 2 \times (3 + 5)^2$
2. **Step 1** $3 + 5 = 8$
3. **Step 2** $2 \times 8 = 16$
4. **Step 3** $16^2 = 256$, so $4 + 256 = 260$

**Which step has the mistake? Explain what went wrong, then show the correct work.**

CORRECT WORK

**Correct answer:** _____

**3** Answer in complete sentences.

Seven friends split the bill for a pizza party. They order 4 pizzas at $3^2$ dollars each, plus a one-time $6 delivery fee. Write ONE numerical expression for what each friend owes, evaluate it, and explain why the grouping symbols matter.

WRITE YOUR ANSWER

EXPLAIN YOUR THINKING

**Two students evaluate $3 + 4 \times 2$ and get 14 and 11. Which one followed the order of operations, and why does the order have to be agreed on at all?**

**3-Point Exit Checklist:** ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

**How did it go?** ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help